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Life-of-Well Approach for Optimizing Production – GOM Case Histories

Eduardo Proano*, Yogesh Patel**, Roxy Thawer**, Salah Al-Harthy*, Bala Gadiyar*, Colin John*, Hugo Morales*

* Schlumberger Oilfield Services, ** BP America. All SPE Members.

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Abstract

A life-of-well comprehensive methodology for production optimization is demonstrated through two Gulf of Mexico stimulation treatments in formations requiring sand control. The approach considers the full well history in the process of well evaluation and candidate selection. The treatments were designed for remediation of damage caused by a time-dependent disruption of formation sand and/or sand-control mechanism, including fines migration, proppant instability, and scaling.

Traditionally, data from permanent bottomhole gauges have been used as a valuable tool for well and reservoir management. As demonstrated in this paper, the data can be crucial to the assessment of the evolution of time-dependent skin. The practical result of well or reservoir surveillance and diagnosis from permanent gauge or traditional production data is shown to produce better decisions on stimulation and production management, while maintaining life-long well integrity.

Introduction

Completion optimization for sand control has been discussed in the literature by several authors.^{1,2} This paper deals with the identification and the remedial actions to reduce or remove completion damage associated with sand control for wells with and without permanent downhole pressure gauges (PDGs). With the balance required between the value of information and the overall economics of a development and exploitation project, it becomes extremely important to capture all events during the life of a well and to utilize all information to the fullest extent.

The use of high-frequency data or continuous information has a great impact in the expansion of the traditional time scale, from the usual snapshot approach to a continuous

evaluation and remedial action process. This process can lead to real-time production management and decision making.

The integration of historical well information from different sources and continuous or high-frequency data, and their availability for practical use through proper evaluation tools, remains a challenge. However, the lessons learned are quickly developing a knowledge base that can greatly impact well and reservoir management in the future.

Life-of-Well Approach

Several factors play significant roles when dealing with completion decisions, especially in the offshore and deepwater environments. The feasibility for any development project rests not only on available reserves but also on the longevity and integrity of well completions. Critical early time and baseline well fingerprinting information is sometimes not available, or not recorded, which can significantly impact the decision-making process on all downstream operations. This situation is equally applicable to single- and multiwell cases.

Completions requiring sand control are especially prone to time-dependent degradation due to the evolving formation and proppant mechanical conditions, which at the end impairs hydrocarbon production through the gravel pack and/or frac-pack elements. It is extremely important to be able to forecast realistic dynamic operating well conditions when planning a sand-control completion. Planning enables the selection of the most appropriate completion method, including the corresponding hardware configuration and job execution program to achieve the expected results.

Fig. 1 presents the life-of-well approach for completion and production optimization. This continuous, perennial process is initiated at the early exploratory and development stages and continues until depletion and abandonment. Below is a brief description of the basic elements of this approach.

WCP Forecast. The life-of-well process begins with the well completion productivity (WCP) prediction that uses all reservoir static and dynamic information. The data are fed into the reservoir module of the predictive model, which will also include information about heterogeneity, lateral extent, and expected drive mechanism. In unconsolidated formations and/or those with stress-dependent porosity and permeability, a sand prediction model and a geomechanical model may be needed on a sandface or regional basis.

Subsequently, all sandface and upper (tubular) completion elements are added to the model. Realistic operational

boundary conditions such as sea floor or platform backpressure are used at the predictive stage

The net result of the productivity forecast is the definition of a target rate through basic pressure and flow rate decline functions. The forecast will yield ranges of initial production and pressure and production decline with the parameters under consideration. The final completion decision and target production rate can be fine-tuned through sensitivity analysis.

Completion Design. Based on the target rate from the forecast, a detailed completion design is put into place that includes all hardware components, perforating program, fracture calibration tests, and gravel-pack or frac-pack placement pumping schedule. Hardware components that provide sand control and fluid loss control are included based on the project. Well monitoring sensors, permanent downhole gauges (PDG's) and their vertical positions, density meters, multiphase flow meters, and fiber optics for distributive temperature measurements are part of the well completion equipment. These components, together with intelligent completion systems, can provide the best tools for real-time production management.

Job Execution. The execution of the completion should incorporate provisions to capture information that can confirm if the completion design has been met and will help to define baseline completion and reservoir parameters.

Pressure measurements made while perforating can yield slug or conventional transient test signals that can be used to estimate transmissibility, kh/μ . An impulse test³ can also be performed while perforating, provided careful operational procedures are followed to generate a valid pressure signal to estimate kh/μ .

Live annulus information, surface and downhole pressure and temperature recorders, and pumping rate history provide critical information to verify the success of the job execution. This information can also be helpful for the subsequent well productivity evaluation. Pressure and injection rate data from pumping and fracture calibration tests can be used to estimate kh/μ from falloff and post-closure analysis.⁴ The establishment of this baseline from early transmissibility evaluation may be critical for later well productivity evaluation.

Well Productivity Evaluation. This phase is probably the most important of the life-of-well approach. In essence, it allows personnel to do the following:

- Select and utilize all available data from Day 1.
- Validate events and calibrate the WCP model to identify time-dependent skin and other parameters affecting production rate.
- Provide a decision-making tool for well stimulation and timing.
- Use real-time production management while maintaining well integrity.

The first step in the evaluation process consists of a thorough review of the calibration and sand-control pumping sequence including all recorded pressures, pumping rates, and temperature measurements. Achieving a tip screenout on frac-

pack applications, for example, normally results in increased well performance. But such performance may not be long-lasting if critical drawdown conditions are reached. If there is substantial interaction between layers when more than one interval has been completed, a review of the temperature recorder data run during job execution may reveal which interval took more fluid and/or proppant at the various stages of calibration and/or sand-control completion pumping. This information would be a direct indication on how the different intervals would behave once the well is put on production.

Production management should be in accordance with critical drawdown considerations, based on the evaluation of formation mechanical properties and/or a sand prediction model. It is not an uncommon occurrence for production rate to decline at a fast rate once critical drawdown conditions have been reached. The well productivity evaluation using the appropriate baseline kh/μ would help anticipate the onset of this milestone and would provide a better roadmap for future drawdown management and specific remedial action.

Ideally, a baseline production profile from production logging or fiber optic temperature measurements can also help to define an early well fingerprint.

Regardless of the available data and of the source to establish baseline kh/μ , the well productivity evaluation model is ideally the same as the model used at the early stages of the productivity forecasting. The evaluation model will be continuously updated as knowledge of the complete well production system is increased.

Key Considerations for Productivity Evaluation

As use of the life-of-well process increases, the previous approach of taking snapshots for well productivity evaluation is quickly changing to an enlarged, continuous time scale. The resulting expanded vision during the life of a well brings a totally different perspective into the well evaluation process.

Wells with PDGs. In our experience from one of the cases presented in this paper, the applications of permanent downhole gauges range from practical uses for well operations to detailed analysis for detection of production impairment, decisions for well stimulation, and continuous production management.

Some key considerations to maximizing gauge data should include:

- **Data Management and Evaluation tools** Permanently installed sensors generate high-frequency data and the amount of information can be overwhelming for practical use. Accordingly, data management and evaluation workflow become a critical aspect of the life-of-well process.
- **Capture of All Events.** Because of an expanded, continuous time scale brought forth by the high-frequency data, events in a life-of-well process cannot be treated as discrete. Each event should be captured and documented because it may impact future actions and decisions.
- **Vertical Position of the Permanent Downhole Gauge.** Due to vertical position of a PDG in typical sand-control completion, analysis of the data can be complex due to

the difficulty in continuously extrapolating to datum depth under dynamic and static conditions.

- **Flow Rate Reconstruction.** In a sand-control environment, time-dependent skin cannot be diagnosed or differentiated from rate-dependent skin without a good flow rate reconstruction process. Time-dependent skin increases for a corresponding decrease in flow rates as frac- and gravel-pack degradation takes place. **Fig. 2** illustrates the process of time-dependent skin identification using the Δp and $\Delta p'$ normalized log-log plot with the pressure derivative.⁵

Wells without PDGs. When dealing with traditional production data, the biggest hurdle is the ability to decompose the formation flow capacity, kh , and skin values that are compounded in the calculated or measured productivity parameters. Production loss is seldom affordable, and buildup tests are not commonly run to monitor the life of a well.

Some of the key points discussed for the PDG case are also applicable here, but the following considerations should be made for well productivity evaluation of traditional data:

- **Early Transmissibility and Job Execution Evaluation.** The best chance to obtain a baseline kh/μ is post-closure analysis from injectivity calibration test pumping and pressure data. The rate signal generated should be compatible with the analysis model so that the inverse problem can be solved. Accordingly, properly designed and executed tests are critical.
- **Develop and Use a Pipe Model.** We use the term pipe model to represent a suitable vertical lift performance model that calculates flowing bottomhole pressure (BHP) from traditional data.
- **Using the WCP Model with Production Systems Analysis.** Once kh and BHP are available from the previous steps, kh and skin can be decomposed based on production systems analysis. The productivity model is then used to history-match past production performance. As skin evolution is monitored in wells requiring sand control, remedial action can be planned and executed.

Well 1 Case History

Well 1 was completed in two zones, lower and upper, requiring sand control, with a PDG in place. The well and reservoir parameters are listed in **Table 1**.

The gas production and BHP history at gauge depth are shown on **Fig. 3**. Originally, the well was put on production from the lower zone in July 2001. Subsequently, the well produced from the upper zone only from late January to mid March 2002. Both zones were placed on commingled production until early May 2002. At this time the first acid stimulation job was conducted on the lower zone, which was now the only interval open to production. A second acid treatment was performed on the lower zone in early October 2002. This zone continued to produce until late November 2002, when both zones were put back on commingled flow.

This paper analyzes the production for only the lower zone until November 2002. Two different segments are considered in accordance to the flow history, Pre-Acid and

Post-Acid production as labeled on **Fig. 3**. All results and conclusions are based on analysis performed at PDG datum.

Pre-production. The WCP model forecast, design, and evaluation phases were run in accordance with the previously described process.

Logging information was acquired during drilling and an openhole logging program was run followed by a modular dynamic test and fluid sampling. The logging program included a dipole sonic induction log, required to optimize the perforating and overall completion strategy.

Static and dynamic parameters were entered into the WCP model that included sandface completion and tubular components to estimate initial production under the expected operational boundary conditions. All hardware was designed to cover production from both zones. A PDG was planned as part of the completion string to monitor BHP, located at 10,049 ft measured depth.

A sand prediction model was run for the planned 21-spf big-hole perforating program. The model predicted sand production at virtually zero drawdown; i.e., the critical pressure was very close to the reservoir pressure. A frac-pack completion was designed to control sand production and minimize skin effects. Proppant was transported through shunt tubes in the lower zone to ensure the 92-ft perforated interval was fully packed. Tip screenout was achieved with almost 200,000 lbm of proppant placed in the fracture. Tip screenout was also achieved on the upper zone with approximately 61,000 lbm of proppant. A formation isolation valve was used to prevent loss of zinc bromide into the formation.

Early transmissibility evaluation from post-closure analysis of the calibration test data yielded a value of $kh/\mu = 3,570$ md-ft/cp, as shown on **Fig. 4**. The resulting kh was later confirmed by transient analysis during the pre-acid production period.

As expected from the WCP model forecast, the lower zone tested at approximately 30,000 Mscf/D and 2850 STB/D of condensate, at a flowing tubing pressure (FTP) of 5,400 psig.

Pre-Acid Production Evaluation. **Fig. 3** shows the BHP and gas flow rate evolution for the lower zone pre-acid production segment.

It was evident that volumetric behavior was taking place during the first few months of production. This behavior was confirmed with material balance calculations that yielded a drainage area of approximately 90 acres. The volumetric behavior slowly changed in later stages of pre-acid production due to an apparent aquifer effect.

During the pre-acid period, several unplanned buildup transients took place. These buildup periods (some of which were very short in duration and could not be analyzed) confirmed the early volumetric behavior was due to a lack of pressure recovery and was manifesting increasingly larger pressure drawdown for decreasing production rates. This condition indicated critical drawdown may have been reached very early in the life of the well.

Detailed skin and rate-normalized drawdown calculations confirmed skin was rapidly evolving as a time-dependent variable. This behavior also seemed to confirm the results from the sand prediction model.

The kh value of 2,350 md-ft agreed with that obtained from post-closure analysis. There was no evidence that either permeability or porosity were stress dependent during the pre-acid production segment. The damage mechanism seems to have been caused by a non-catastrophic formation failure mechanism, fines migration into the frac pack. **Fig. 5** shows the skin evolution based on individual buildup period analysis. Clearly, the skin damage mechanism evolved as an exponential function of production time. It is estimated that the total equivalent skin value reached 195 by the time the lower zone was shut-in (late January 2002).

The exponential increase in skin was confirmed by KPIs. **Fig. 6** shows the rate-normalized pressure drawdown and corresponding flow rate for several individual buildup periods, using the instantaneous reconstructed flow rate prior to shut-in (the analysis was conducted with pseudo-pressures, but for simplicity the idea is illustrated in terms of pressure). A correlation was found for a rate-normalized pressure function based on p_{1hr} , p^* , p_{last} , and p_r from the history-match model, as this function showed the same behavior for all pressure bases. This correlation may be useful for PDG application in a sand-control environment, because it may give an early warning signal and continuous measure of skin evolution when critical drawdown conditions have been reached.

The key point about this analysis is that for time-dependent skin evolution the flow rate is inversely proportional to the rate-normalized pressure drop. This is contrary to the rate-dependent skin case, in which flow rate is directly proportional to it. Additionally, while the rate-normalized pressure drop may increase in a nonlinear form, the nonlinearity may not be completely attributed to the nature of time-dependent or rate-dependent skin, but may be the result of gas flow behavior.

Because of the high volume of unplanned shut-ins in some Gulf of Mexico operations, setting a reference pressure to perform the rate-normalized function may be useful on a field-by-field basis.

History Matching. The first attempt for simultaneous transient analysis of qualified periods and history matching made use of a numerical multiphase model. The well is the only producer of the target zones, and it is near two intersecting faults with the gas/water contact on the down-dip side. The model worked reasonably well during the first few weeks of production while volumetric behavior was predominant. However, because the water onset occurred unexpectedly and the skin evolved exponentially, it was possible to replicate only the macroscopic behavior. It was almost impossible, as well as impractical, to history match high-frequency data and confirm the microscopic events that characterized the sandface condition.

An analog single-phase gas analytical WCP model, which reasonably duplicated the flow regimes resulting from geological features, provided a reasonable material balance control and also allowed the implementation of a changing well model, using superposition as an approximation to handle the time-dependent skin evolution. This model was used because we had good material balance control, we could monitor the instantaneous skin condition affecting well productivity, and the evaluation process would allow us to

take action for real-time production management without the normal reservoir management and development constraints typical to a multiwell system. **Fig. 7** shows the history match model results using the variable skin, changing well model, based on skin values calculated from qualified buildup periods (**Fig. 5**) during the first 185 days (~4500 hr) of production from the lower zone.

Upper Zone and Commingled Production. Upper zone and subsequent commingled lower zone flow took place between late January and early May 2002 (**Fig. 3**). During this time the lower zone was further depleted. Material balance was adjusted based on the value of reservoir pressure before acid. Unfortunately, it was not possible to track the skin evolution on the lower zone sandface during this period, but it is likely that it continued to increase above 195, the last calculated value on late January 2002. The lower zone was tested at 4,713 Mscf/D and 213 STB/D of condensate at a FTP of 843 psig.

Acid Stimulation Treatments. For practical reasons, both acid stimulation treatments are discussed in this section, rather than presenting them in the chronology of the paper. The increase in skin was attributed to three main factors:

1. Fines migration: Based on sand strength analysis, between 5 and 30% of the clays and silts could migrate under suitable conditions. There was onset of water production, which coupled with high gas production rate could trigger fines migration. Fines could be in the formation, proppant pack, and screen.
2. Inorganic scale: Analysis of produced formation water after the first acid treatment indicated a moderate probability of calcium carbonate (CaCO_3) scale buildup under bottomhole conditions.
3. Condensate block: The reservoir was in the depletion stage, so gas was assumed to be below the dew point that could result in condensate block in the formation.

First Acid Treatment. To address fines migration, a combination of hydrofluoric acid (HF) and clay acid systems were chosen. The HF system would aid in partially dissolving the fines, whereas clay acid would stabilize undissolved fines, thereby making them immobile. Because of the limited knowledge of mineralogy and the high temperature, it was decided to formulate the HF and clay acid with an organic acid. In addition, to prevent water block phenomena and aid in flowback, 30% methanol was added to all the stages. The diverter was 70 quality nitrogen foam. A total of three acid stages (two preflushes and a main acid) and two diverter stages were proposed. Following is the treatment sequence:

- Preflush 1: 50% methanol and surfactant
- Preflush 2: 10% acetic acid, corrosion inhibitor, methanol, and surfactant
- Main acid: Organic mud acid, corrosion inhibitor, methanol, surfactant, and iron control
- Diverter: Nitrogen foam
- Overflush acid: Organic clay acid, corrosion inhibitor, methanol, and surfactant

The treatment was executed by a stimulation marine vessel and was bullheaded down the production tubing. The rates varied from 0.5 to 3.0 bbl/min. **Fig. 8** illustrates the treatment plot. Good breakdown and diversion were observed when acid and diverter stages were at the perforations, respectively.

Second Acid Treatment. This treatment was designed to address fines migration and mitigate CaCO_3 scaling. The major changes were a) addition of acid-compatible scale inhibitor to preflush acid, b) change to HCl-based acid systems including clay acid, c) addition of HCl overflush stage, and d) increase of the acid coverage (gal/ft) by minimum of 30%. Following is the treatment sequence:

- Preflush 1: 50% methanol and surfactant
- Preflush 2: 7.5% HCl, scale inhibitor, corrosion inhibitor, methanol, surfactant, and iron control
- Main acid: Mud acid (HCl:HF), corrosion inhibitor, methanol, surfactant, and iron control
- Overflush acid 1: 5% HCl, corrosion inhibitor, methanol, surfactant, and iron control
- Diverter: Nitrogen foam
- Overflush acid 2: Regular clay acid, corrosion inhibitor, methanol, surfactant, and iron control.

The treatment execution was similar to the first treatment. Produced water samples were requested at regular time intervals to monitor activity of scale inhibitor in produced water. **Fig. 9** demonstrates that scale inhibitor concentration is still above minimum inhibitor concentration (MIC), indicating the well is still protected.

Post-Acid Production Evaluation. Here again, for practical reasons, both acid stimulation treatments are discussed in this section. **Fig. 10** shows the post-acid production profile. Clearly the first acid stimulation treatment in early May 2002 was very successful: the well produced at 14,649 Mscf/D with 905 STB/D of condensate at a FTP of 1,337 psig 1 week later.

None of the unplanned shut-in periods following the first acid stimulation (see **Fig. 3** for pressure history) were valid for transient analysis. However, use of the rate-normalized KPI for skin evolution analysis determined that the evolution of skin continued to be on an exponential mode. This is shown on **Fig. 11** for qualified buildup periods during the post-first acid production. Production continued to decrease during the next 5 months, as the skin maintained the apparent exponential evolution.

The second acid stimulation treatment was successfully applied in mid-October 2002. Production rate increased from 2,660 Mscf/D with 152 STB/D of condensate at a FTP of 824 psig, to 8540 Mscf/D with 530 STB/D of condensate at a FTP of 1,007 psig.

It is possible that during the post-acid production periods the permeability and/or porosity values may not have remained constant. However, this could not be verified from the previous rate-normalized log-log analysis. Accordingly, it was assumed that kh remained constant at 2,350 md-ft through the whole evaluation period. Using this value and the rate-normalized KPI as an indicator, a history match was performed and yielded the flow rate versus skin evolution

function shown on **Fig. 12**. Because of the assumptions made, we choose to use the term “total equivalent skin.”. One conclusion from this graph is that the prevailing sandface damage mechanism exhibits a definite hysteresis; i.e., every acid treatment does not bring the apparent fines problem to its original condition because some residual skin condition is left in place.

Pipe Model. **Fig. 13** shows the DPG and FTP data for the complete production period covered in this paper. In this graph, the calculated BHP at gauge depth has been superimposed. The pipe model for BHP calculation used the Duns and Ros⁸ correlation, with an adjustment to the friction factor. We show that a good pipe model can help identify macroscopic features such as the volumetric behavior and the possible onset of the aquifer support. Most important historical features during flow are reasonably duplicated. The study of the FTP and calculated BHP or BHP data obtained from DPGs shows that good quality, continuous FTP information can also be valuable in recognizing reservoir and sandface behavior. Undoubtedly, for a well with similar characteristics, such as No. 1, the historical FTP data clearly follow the same trend as the BHP at PDG depth, except that it is translated by the prevalent flowing pressure gradient. FTP alone may have given an indication of volumetric behavior and the later changes in drive mechanism and increasing skin conditions.

One important point about the pipe model is its usefulness for flow rate reconstruction and as a check on back-allocation practices.

In the context of productivity evaluation using the principles of production analysis, a knowledge of kh/μ from the calibration tests before completion together with a good pipe model may provide the tools for life-of-well evaluation when continuous BHP data are not available from PDGs. We show the concept on the next case history.

Case History – Well No. 2

The second case history relates to a well completed in one zone with several producing lobes requiring sand control, without a PDG in the completion string. The detailed parameters are listed on **Table 2**. The FTP and production history are shown on **Fig. 14**.

Pre-production. The WCP model forecast, design, and evaluation phases were in accordance with the process, and in a very similar form, as discussed for Well No. 1.

A frac- pack completion system was designed to control sand production and minimize skin effects. Tip screenout was achieved with 153,000 lbm of proppant placed in the fracture.

Post-closure analysis of the calibration test data provided an early evaluation for transmissibility, which yielded a value of $kh/\mu = 3,611$ md-ft/cp, as shown on **Fig. 15**.

This well was put on production in June 2000. As expected from the WCP model, the well tested at approximately 50,000 Mscf/D and 1,520 STB/D of condensate, at a FTP of 4,880 psig.

Pre-acid Production Evaluation. Analysis of the production history for this well rests on a WCP model based on the early kh/μ value obtained from the post-closure analysis of the

calibration tests, together with a pipe model to calculate flowing BHP.

Fig. 16 shows the FTP and gas flow rate evolution with the calculated BHP at PDG depth using a pipe model similar to the one for Well No. 1. The pipe model was calibrated against a production log run prior to the acid stimulation treatment. As the well is put on production, the FTP and calculated BHP enable the monitoring and surveillance of the evolving events. Volumetric behavior during the first few months of production was confirmed with material balance calculations that yielded a drainage area of approximately 420 acres, with an estimated initial gas in place of 57 Bscf. The volumetric behavior seemed to be slowly changing in the later stages of the preacid production from an apparent aquifer support. A cumulative production of approximately 29.1 Bscf was achieved before the acid stimulation performed on April 5, 2003.

This well was completed on a long perforated interval (Table 2), which covered several sand zones with shale sequences. It was assumed for analysis that kh remained constant throughout the production history. The production log recorded before the acid treatment showed that most production came from a very small area close to the middle of the perforated interval. A review of the calibration tests and the job execution showed that the frac-pack coverage extended throughout the perforated interval. Accordingly, the apparent formation failure problem plugged most of the path to flow into the wellbore. However, because of the number of annuli, radial, and vertical proppant-filled or empty paths, the production log measurement point does not necessarily tell that all flow is radial into such point.

A short, 30-min buildup period was available after the production log run, but analysis with the flowmeter rate yielded inconclusive results regarding kh and skin, with a tendency to very small values for both parameters.

History Matching. The gas and condensate production history shows the well exhibited an exponential decline in gas production rate until February 2002, at which time the production rate dropped sharply, coinciding with the first significant increase in water production. Formation failure and frac-pack degradation may have been simultaneous to this event. From this point onward, gas production rate continued to decline at a higher rate. No attempt was made to account for the water production on the reservoir module. The approach taken was to develop an analog well productivity model that would enable a history match of the calculated flowing BHP from the pipe model, and a comparison of the resulting calculated gas rates and cumulative gas production against measured values.

Just as in the previous case, a changing well model was used to account for time-dependent skin evolution. Because of the assumptions made, a total equivalent skin is utilized in the match process. Permeability and porosity reduction cannot be ruled out as a compound effect on the equivalent skin values.

The history match model shows time-dependent skin evolution from the very early stages of production, resulting from a mechanical noncatastrophic failure in the formation once it reached the critical drawdown point. **Fig. 17** shows the final match. The calculated flow rate and cumulative productions based on the pipe model BHP values reasonably

match the measured values. The normalized rate cumulative plot yielded a straight line confirming the initial BHP value, which was 8,124 psia, and the estimated initial gas in place of 57.3 Bscf agrees with the estimated value.

The changing well model with time-dependent skin yielded an estimated value of total equivalent skin of 366 prior to acid stimulation.

Acid Stimulation Treatment. In a 2-yr period, the production rate had declined significantly. The mineralogy data indicated fines, so fines migration was considered the root cause of formation damage. In this case also, to address fines migration, a combination of inorganic mud and clay acids were recommended. Four acid stages (preflushes and main acid) and three diverter stages were proposed. At the end of the job, the well was shut-in for 3 hr to allow clay acid to react with the formation. The treatment sequence was as follows:

- Preflush 1: 50% methanol and surfactant
- Preflush 2: 7.5% HCl, corrosion inhibitor, methanol, surfactant, and iron control
- Main acid: Mud acid (HCl:HF), corrosion inhibitor, methanol, surfactant, and iron control
- Overflush acid 1: 7.5% HCl, corrosion inhibitor, methanol, surfactant, and iron control
- Diverter: Nitrogen foam
- Overflush Acid 2: Regular clay acid, corrosion inhibitor, methanol, surfactant, and iron control.

This treatment was also executed by a stimulation marine vessel and was bullheaded down the production tubing. The rates varied from 0.75 to 3.0 bbl/min and treating pressure ranged from 1,180 to 2,978 psi. Good breakdown and diversion were observed when acid and diverter stages were at the perforations.

Post-Acid Production Evaluation. The acid stimulation treatment in early April 2003 was successful. The well produced at approximately 8,300 Mscf/D with 160 STB/D of condensate at a FTP of 2,350 psig, from previous values of 3,350 Mscf/D and 60 STB/D at 300 psig. Unfortunately, water production jumped from 300 to 1,700 STB/D.

The final history match model for post-acid production is shown on **Fig. 19**. It is estimated that the initial, post-acid total equivalent skin was approximately 35, while the last value is estimated to be at 228. Decline curve analysis shows a gas in place value of approximately 1.8 Bscf, which is quite different from the expected 28 Bscf. The difference may be due to the active aquifer, depletion, or lack of communication to the wellbore from one or more of the individual zones originally perforated. In view of the simplified model used and because the pipe model could only be calibrated against one measured point, no attempt was made to reconcile the discrepancy.

Lessons Learned and Conclusions

- The life-of-well approach is a valuable process for optimizing production from Day 1. This is demonstrated through two GOM cases.
- Forecasting realistic operating well conditions is required when planning a sand-control completion. Proper

planning enables the selection of the most appropriate completion method, including the corresponding hardware configuration and job execution program to achieve the expected results.

- The practice of monitoring and surveillance of events, and continuous diagnosis from permanent gauge or traditional production data is shown to produce better decisions on well stimulation and production management.
- Critical early time and baseline well fingerprinting information can significantly impact the decision-making process on all downstream operations.
- Flow rate reconstruction may be needed to identify time-dependent skin when using PDG data.
- A better understanding of the dynamics of wellbore flow relative to PDG extrapolation and pressure datum selection is required to characterize time-dependent skin evolution.
- When only traditional production data are available, use of the early transmissibility evaluation from calibration tests and a good pipe model may allow post-completion evaluation and remedial action.
- For production management of completions requiring sand control, it is extremely important to have a good reference point for formation failure and critical drawdown with the use of a sand-prediction model.
- The compound effect of skin degradation on gravel- or frac-pack completions with stress-dependent porosity and permeability reduction is not well understood and complicates the time-dependent skin evaluation.

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Nomenclature

k = permeability, md
 Gp = cumulative gas production, Mscf
 h = net pay thickness, ft
 p = pressure, psi
 q_g = gas production rate, Mscf/D
 t = production time, hr or days

Greek symbol

Δ = difference
 μ = viscosity, cp

Subscripts

last = last measured
 r = reservoir
 wh = wellhead
 wf = wellbore, flowing

Superscripts

* = extrapolated

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Table 1 – Well/Reservoir Data for Well No. 1

Zone:	Lower, Sandstone
Perforated Interval (@ 21 spf), ft	11540 - 11632 MD 11050 - 11115 TVD
Net Formation Thickness, ft	58
Porosity	0.195
Water Saturation	0.250
Downhole Pressure Gauge Depth, ft	10049 MD 9866 TVD
Initial Reservoir Pressure, psia	9045 @ mid-perfs 8713 @ gage depth
Reservoir Temperature, °F	235
Tubing Size, in	3.5 O.D., 2.992 I.D.
Casing Size, in	7 O.D., 6.184 I.D.
Completion Type:	Frac Pack
Initial Gas Production Rate, Mscf/D	30000
Initial Oil Production Rate, STB/D	2850 condensate

Table 2 – Well/Reservoir Data for Well No. 2

Zone:	Sandstone
Perforated Interval (@ 21 spf), ft	10430 - 10514 MD 10244 - 10315 TVD
Net Formation Thickness, ft	66
Porosity	0.225
Water Saturation	0.150
Initial Reservoir Pressure, psia	8124 @ mid-perfs
Reservoir Temperature, °F =	202
Tubing Size, in	3.5 O.D., 2.992 I.D.
Casing Size, in	7 O.D., 6.184 I.D.
Completion Type:	Frac Pack
Initial Gas Production Rate, Mscf/D	49000
Initial Oil Production Rate, STB/D	1500 condensate

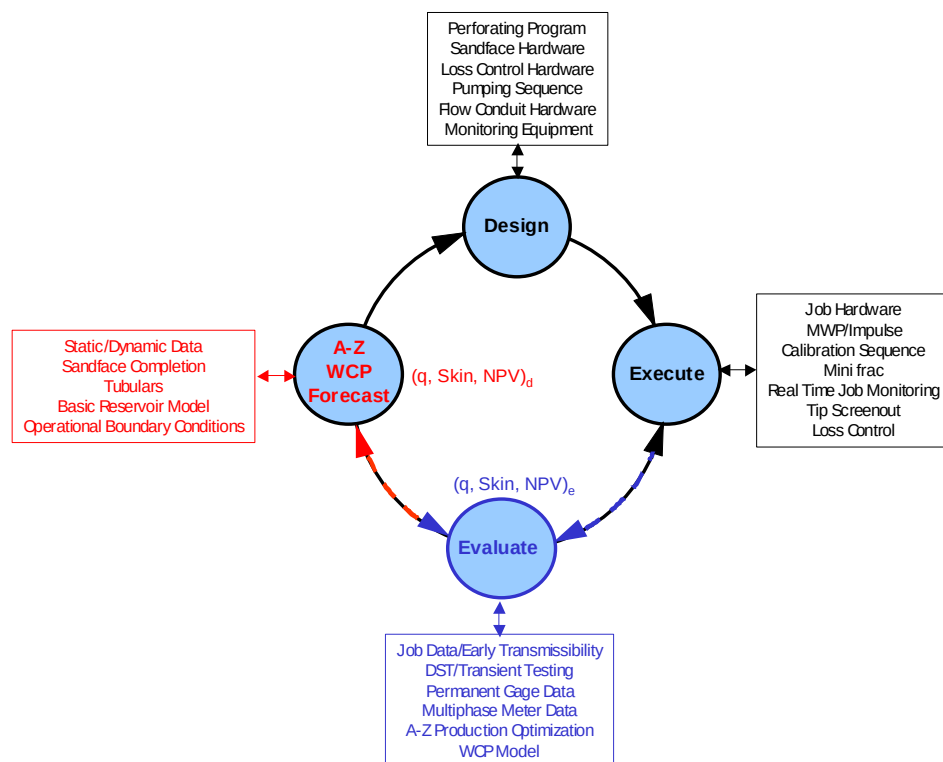


Fig. 1 – Life-of-Well cycle for Completion and Production Optimization

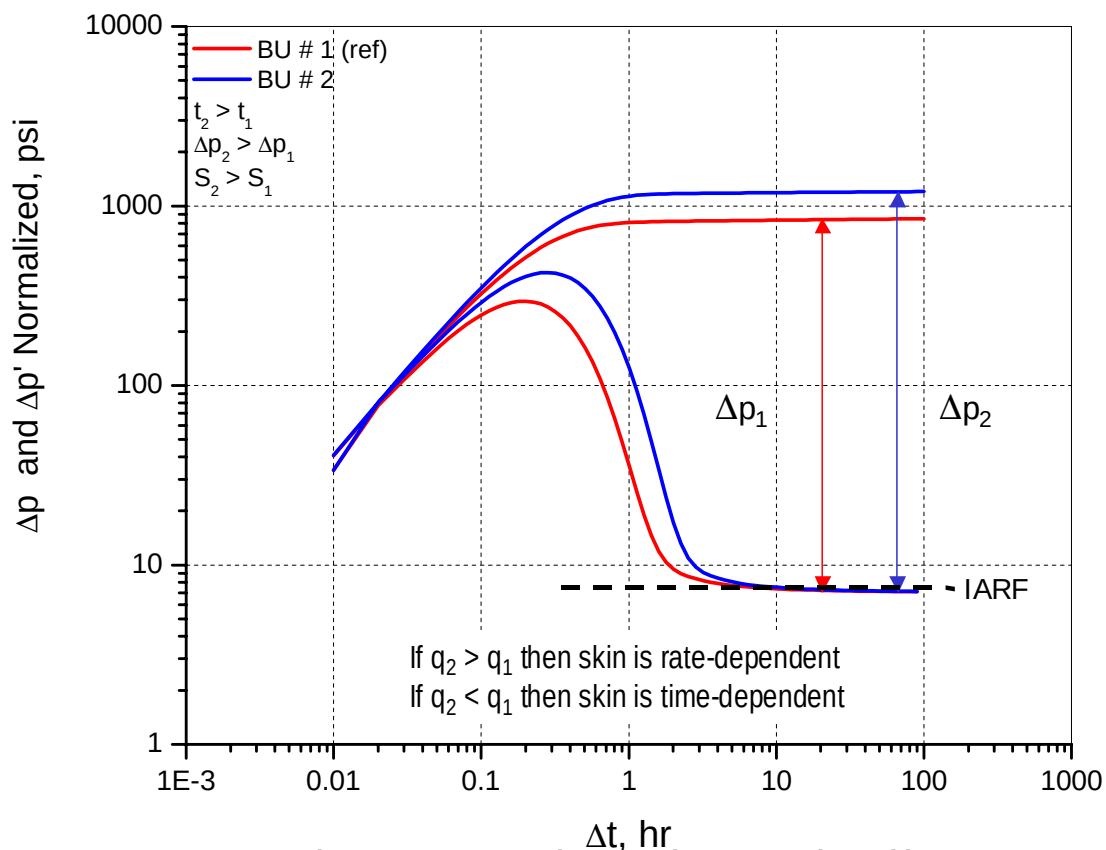


Fig. 2 – Rate-Dependent vs. Time-Dependent Skin

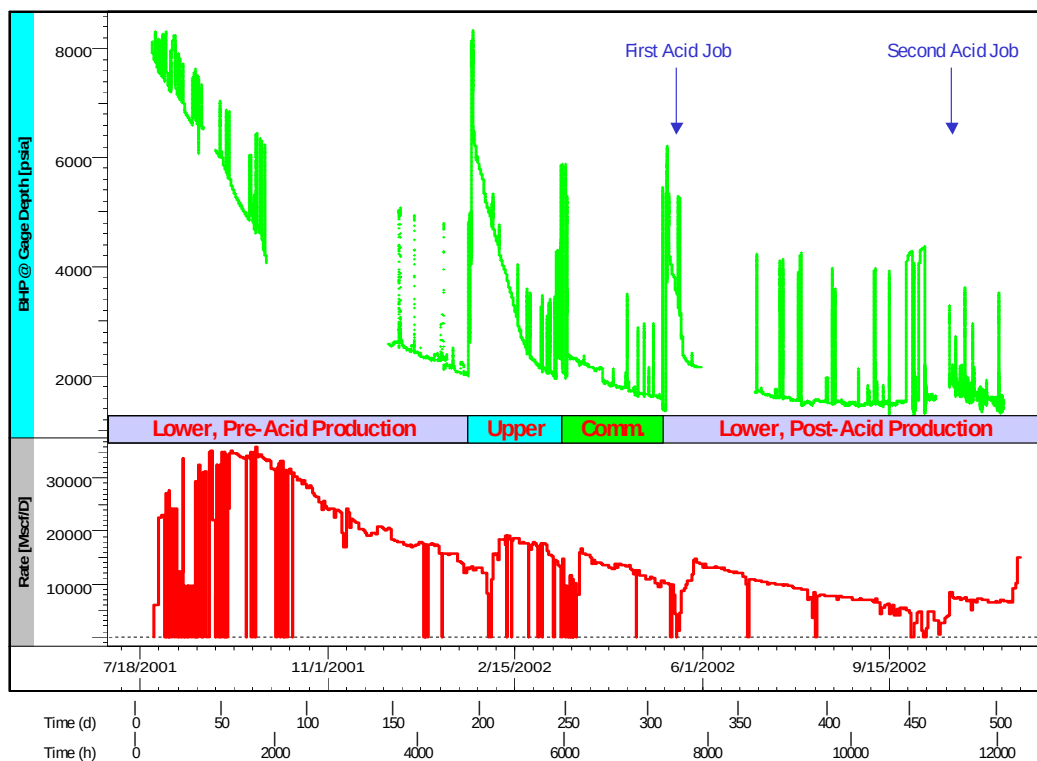
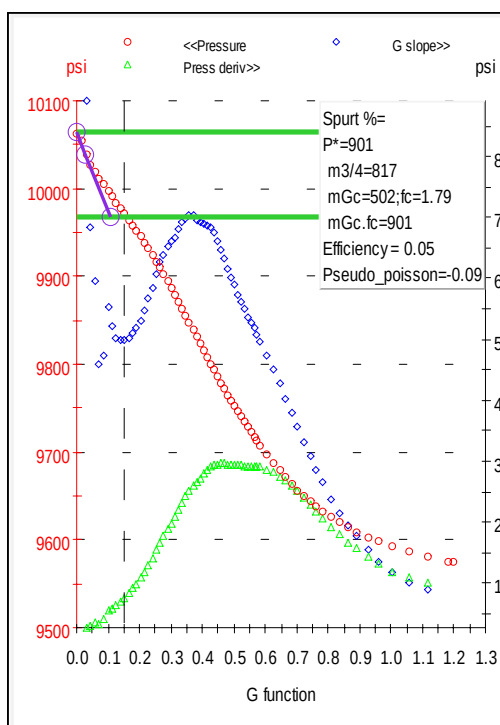


Fig. 3 – Well No. 1 – Pressure and Gas Flow Rate History

G– Function Plot



Flow Identification Plot

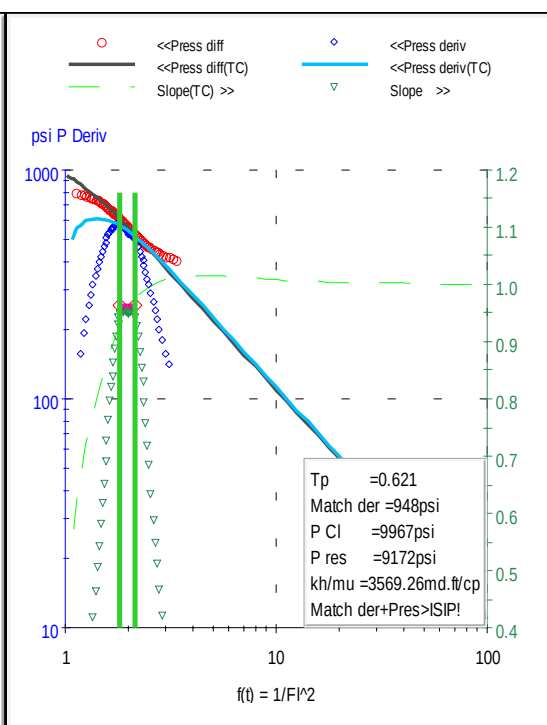


Fig. 4 – Well No. 1 – Calibration Test Post-Closure Analysis

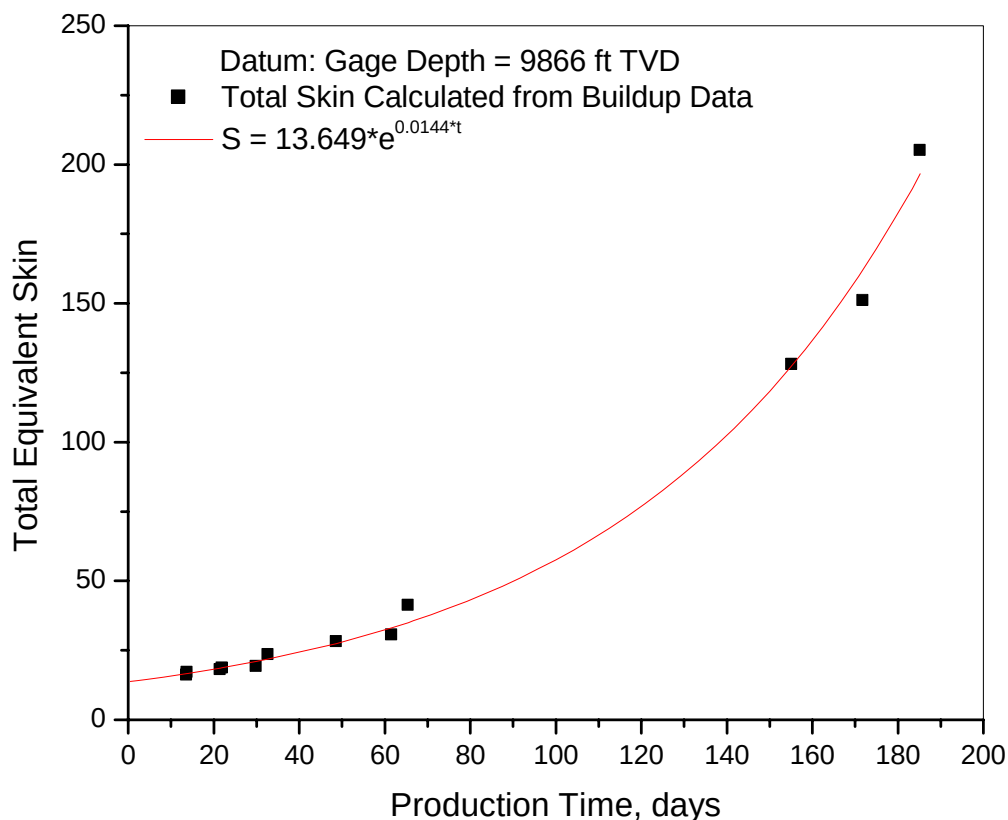


Fig. 5 – Well No. 1 – Evolution of Time-dependent Skin for Lower Pre-Acid Production

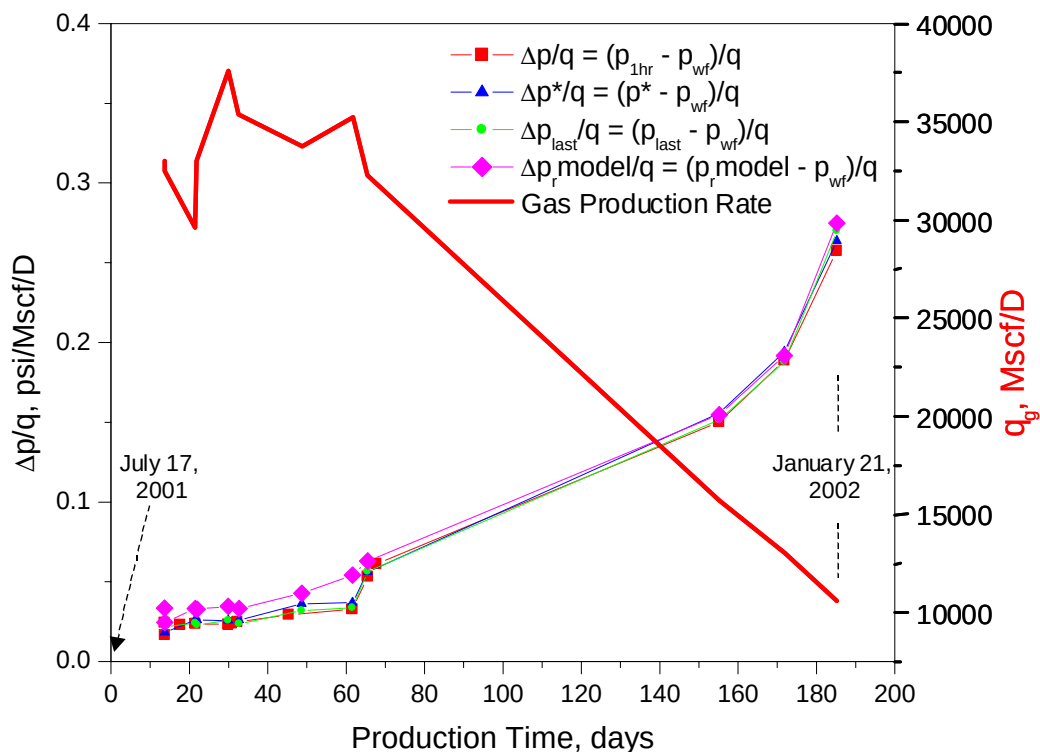


Fig. 6 – Well No. 1 – Rate-Normalized Pressure Drawdown and Gas Rate for Lower Pre-Acid Production

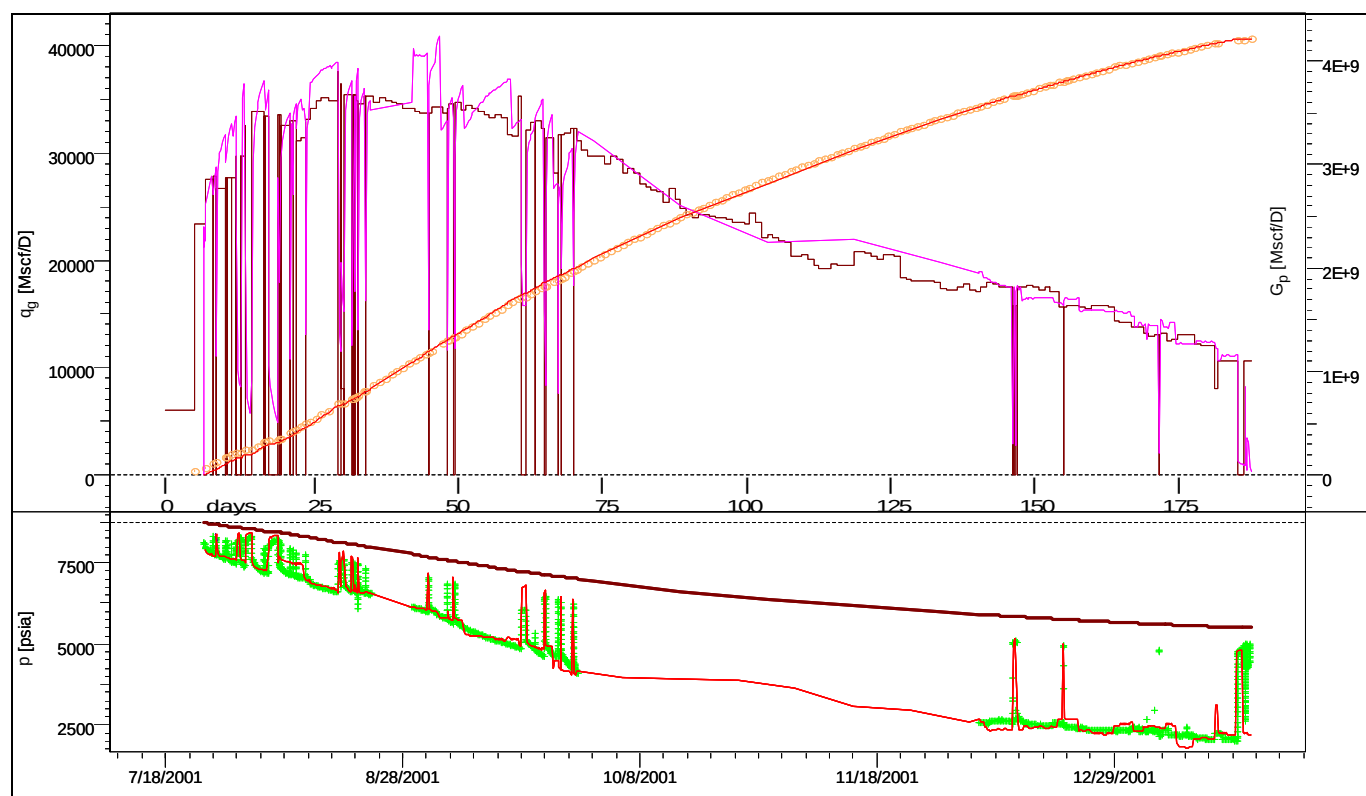


Fig. 7 – Well No. 1 – History Match of Lower Pre-Acid Production

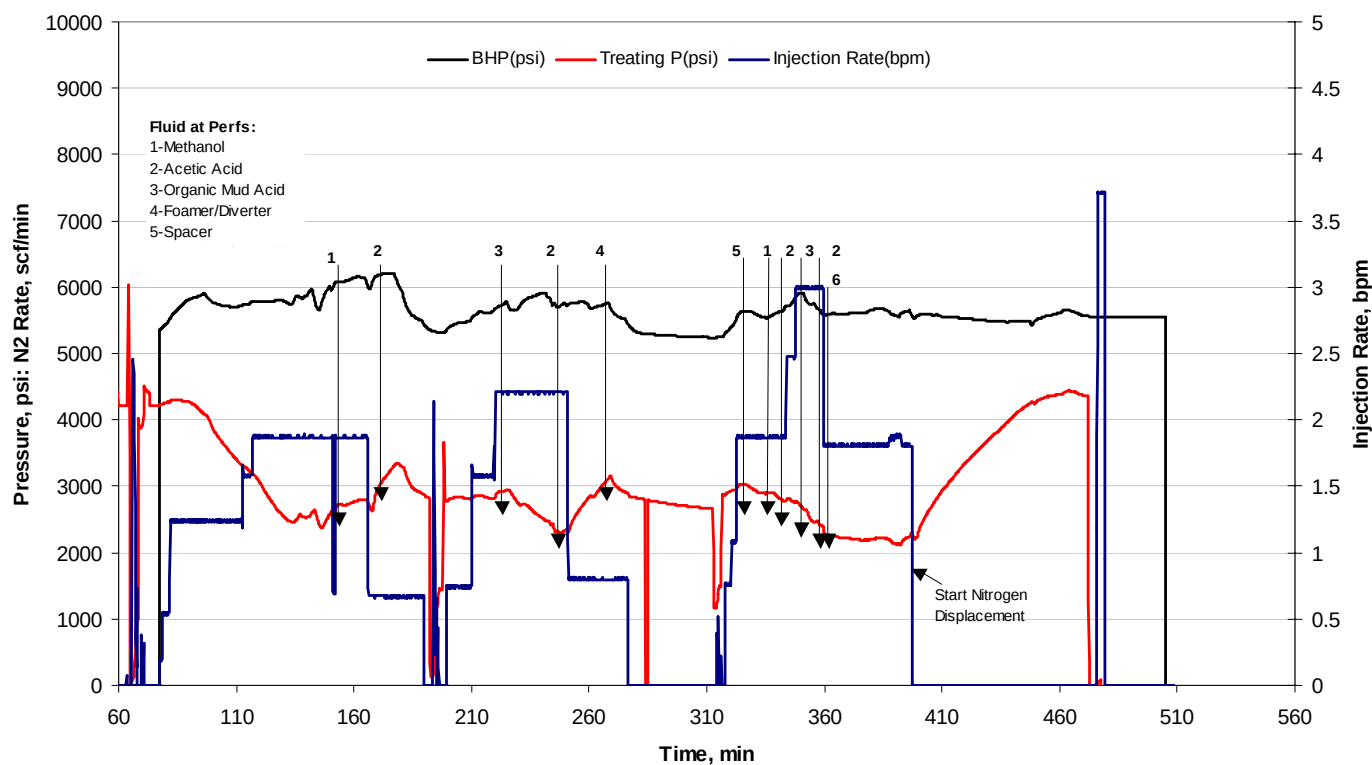


Fig. 8 – Well No. 1 – First Acid Stimulation Treatment

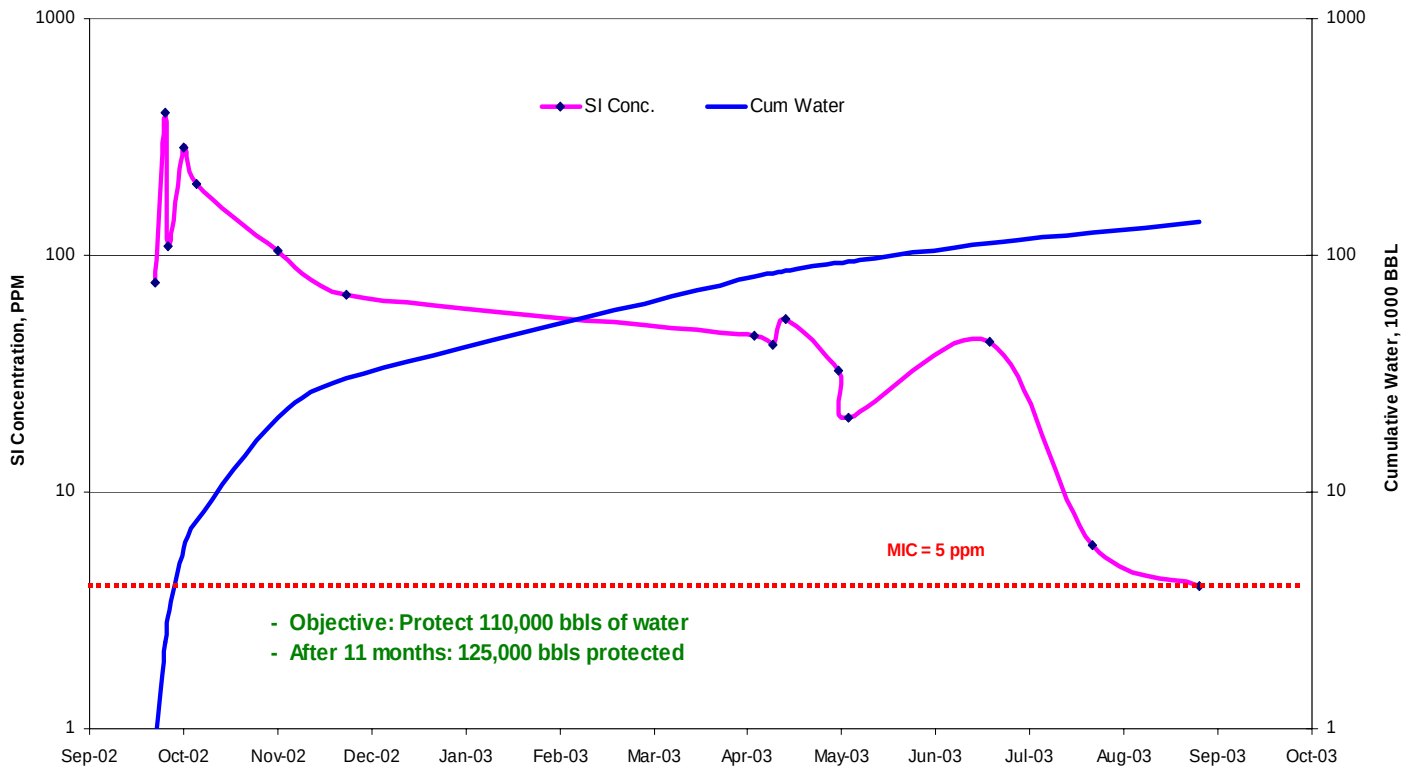


Fig. 9 – Well No. 1 – Scale Inhibitor Return Profile

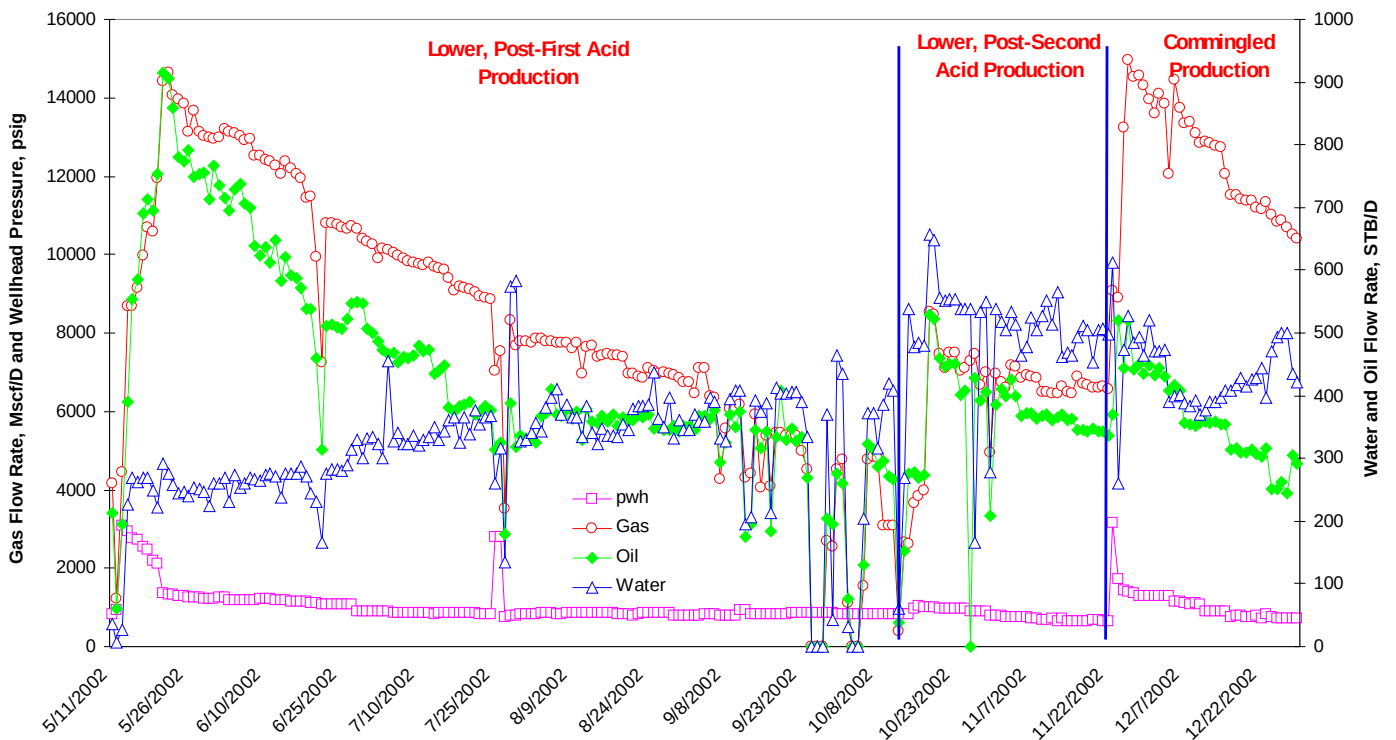


Fig. 10 – Well No. 1 – Post-Acid Production Profile

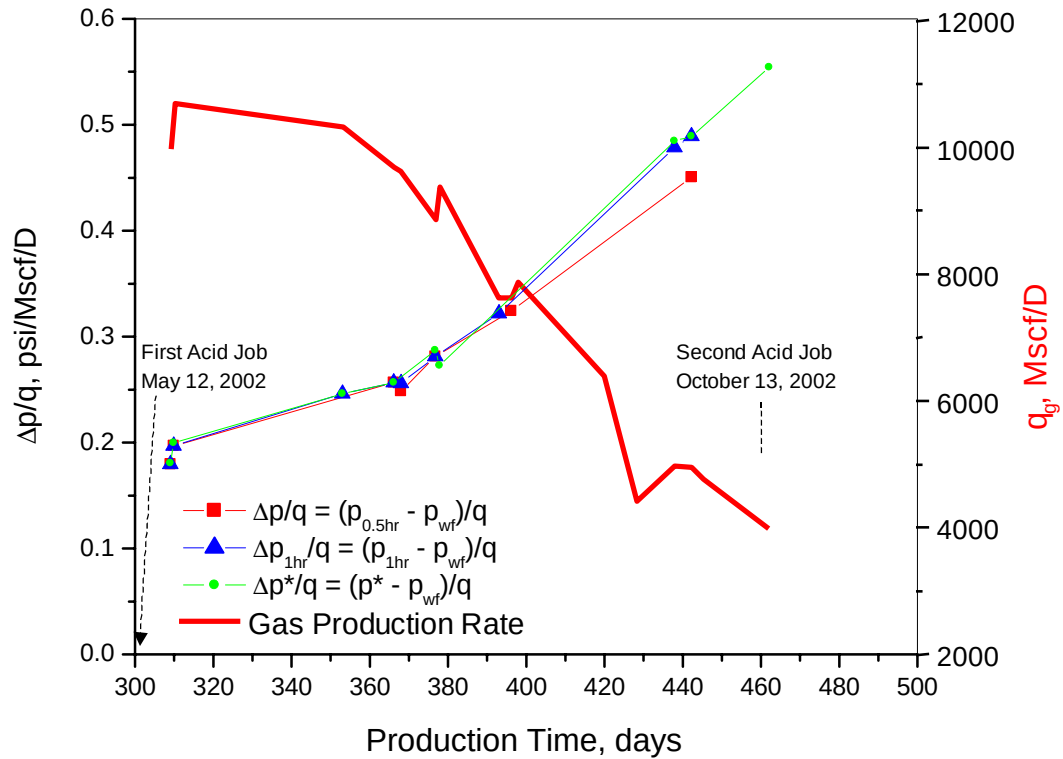


Fig. 11 – Well No. 1 – Rate-Normalized Pressure Drawdown for Lower Post-First Acid Production

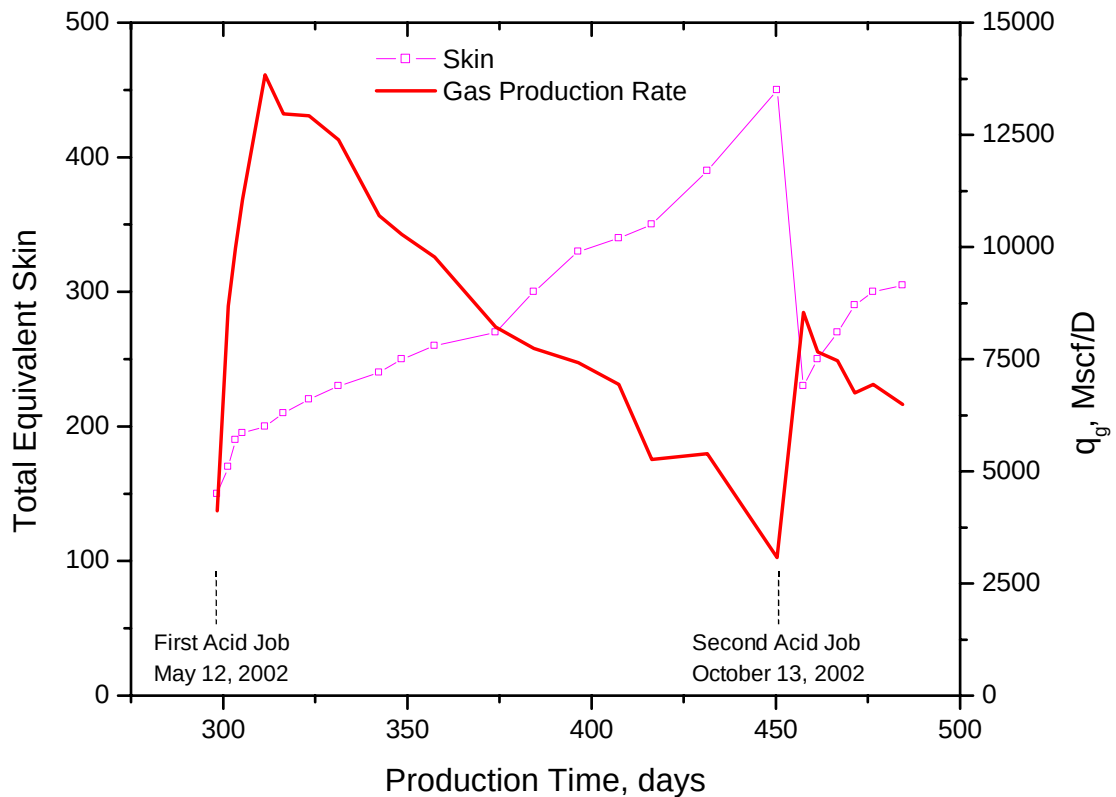


Fig. 12 – Well No. 1 – Evolution of Post-Acid Gas Flow Rate and Skin from History Match Model

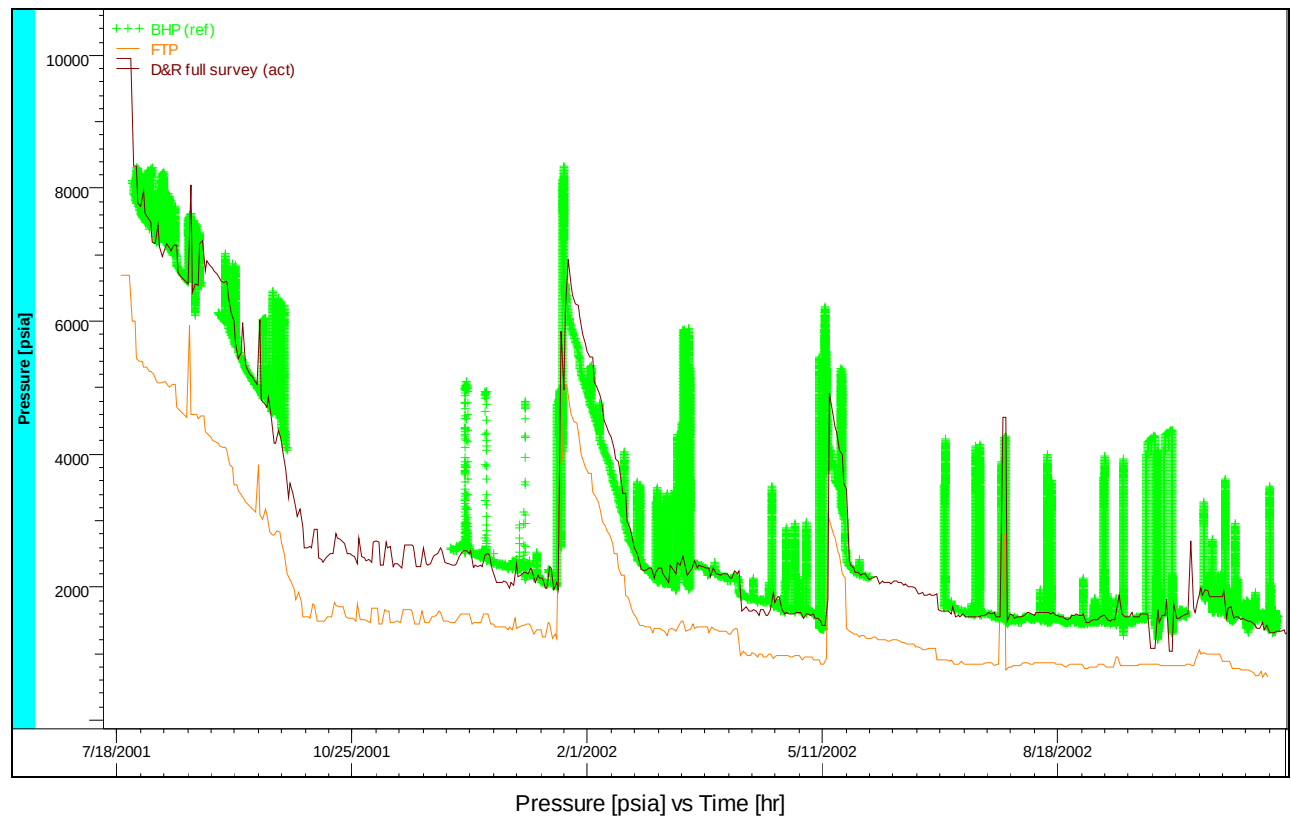


Fig. 13 – Well No. 1 – Pipe Model: FTP, Measured vs. Calculated BHP at PDG Depth

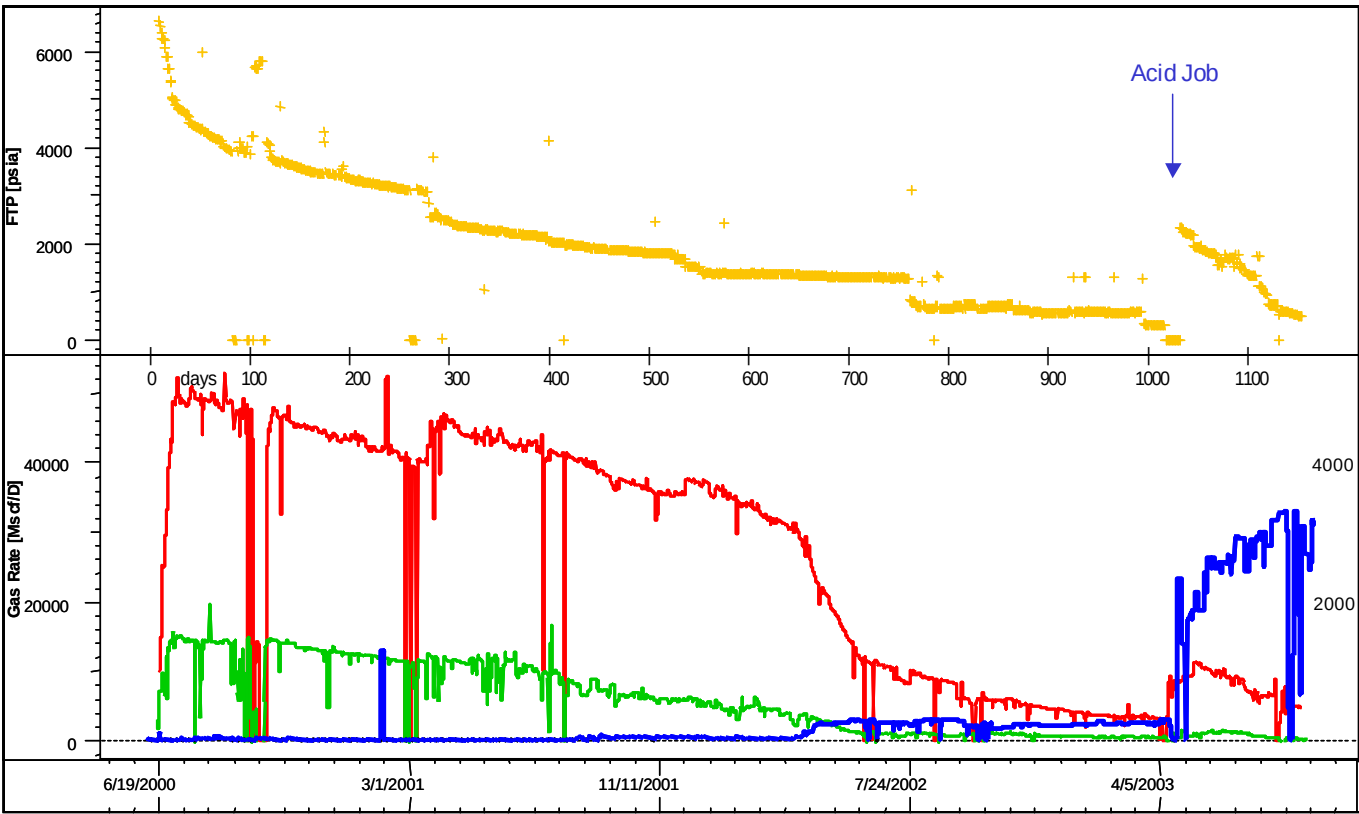
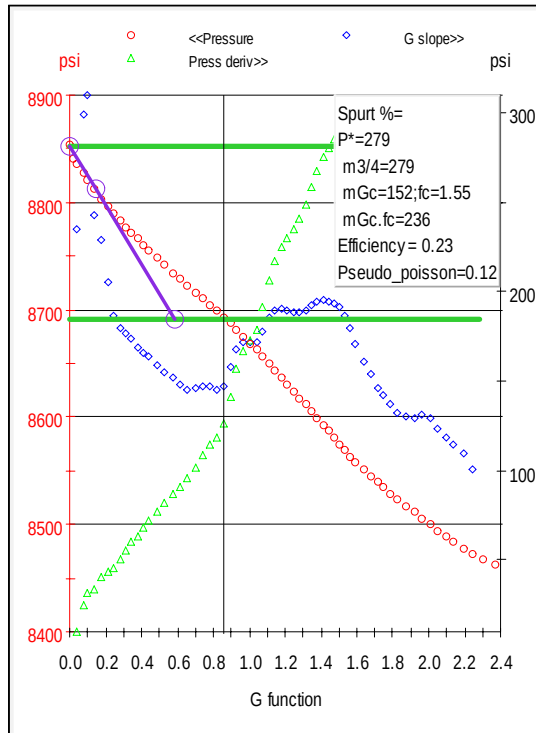


Fig. 14 – Well No. 2 – FTP and Flow Rate History

G– Function Plot



Flow Identification Plot

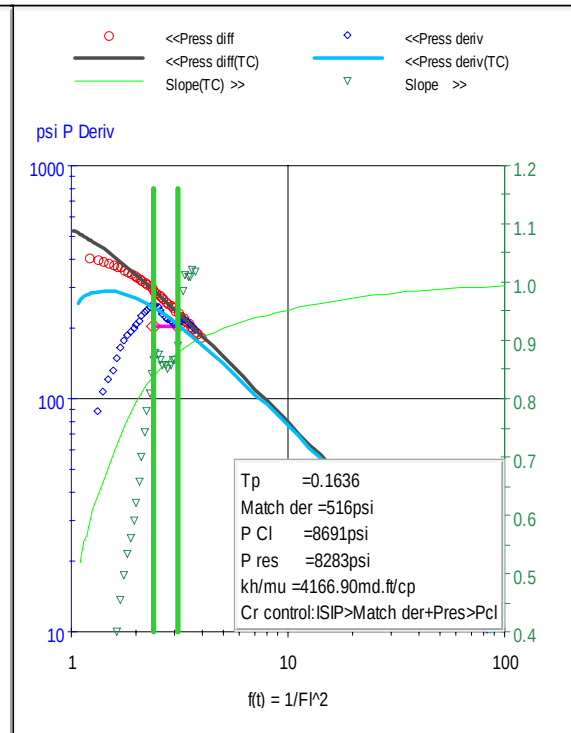


Fig. 15 – Well No. 2 – Calibration Test Post Closure Analysis

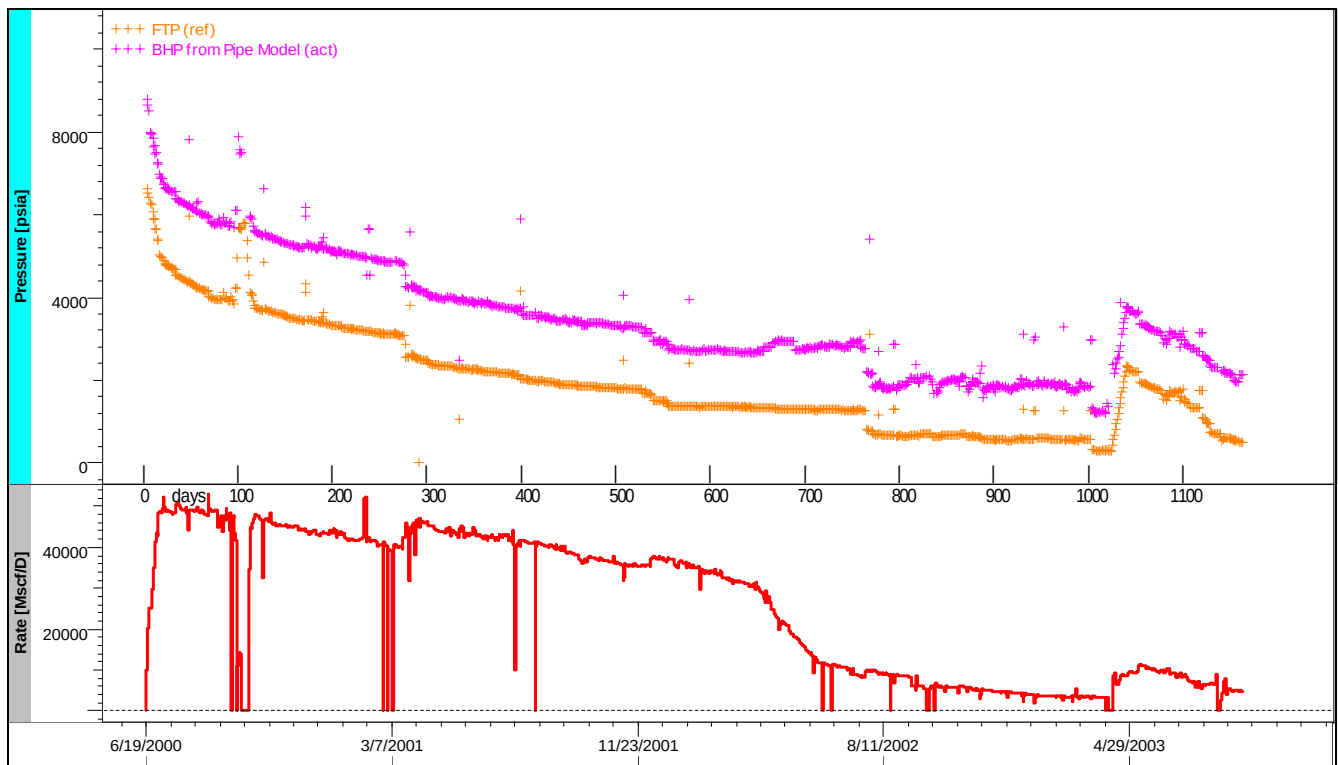


Fig. 16 – Well No. 2 – Gas Flow Rate, FTP and Calculated BHP with Pipe Model

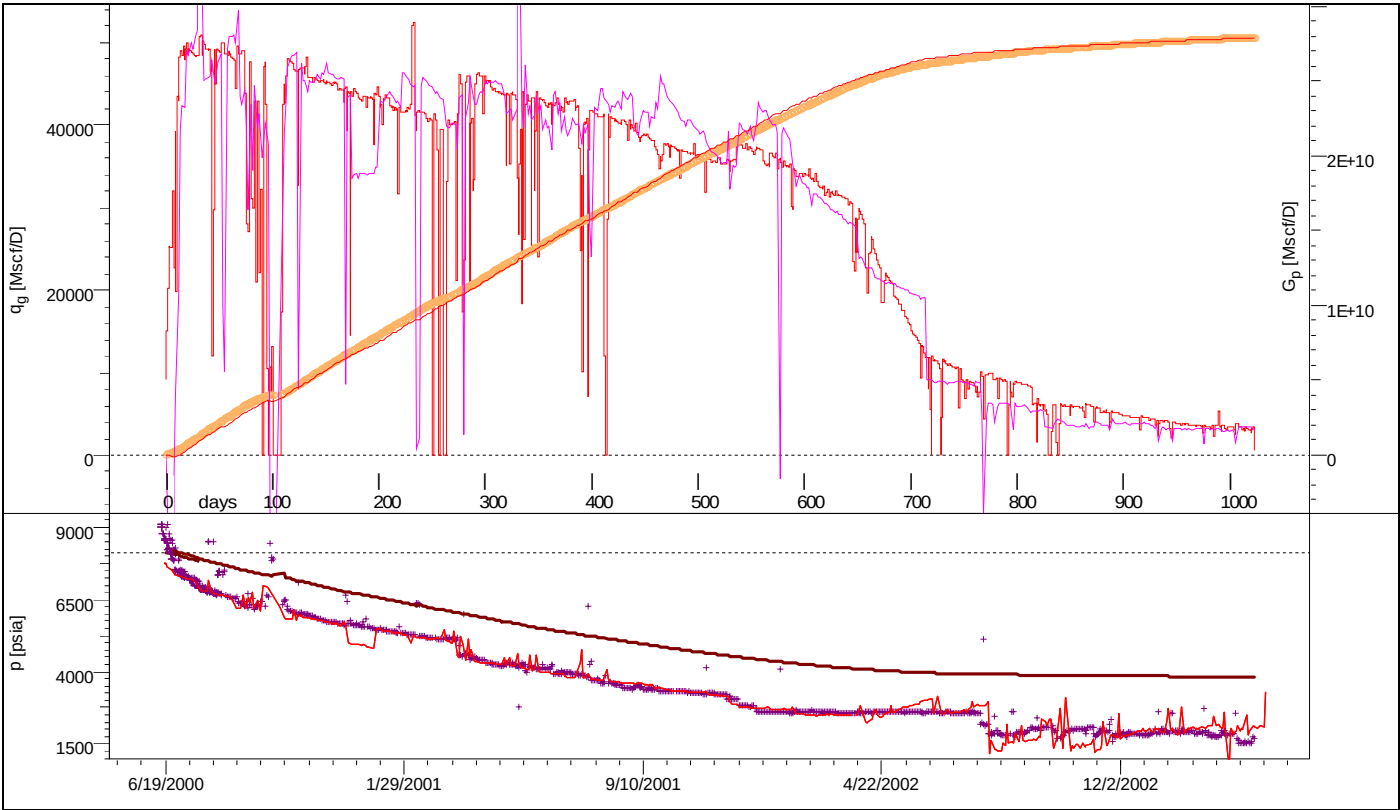


Fig.17 – Well No. 2 – Pre-Acid History Match with the Post-Closure Analysis Results and the Pipe Model

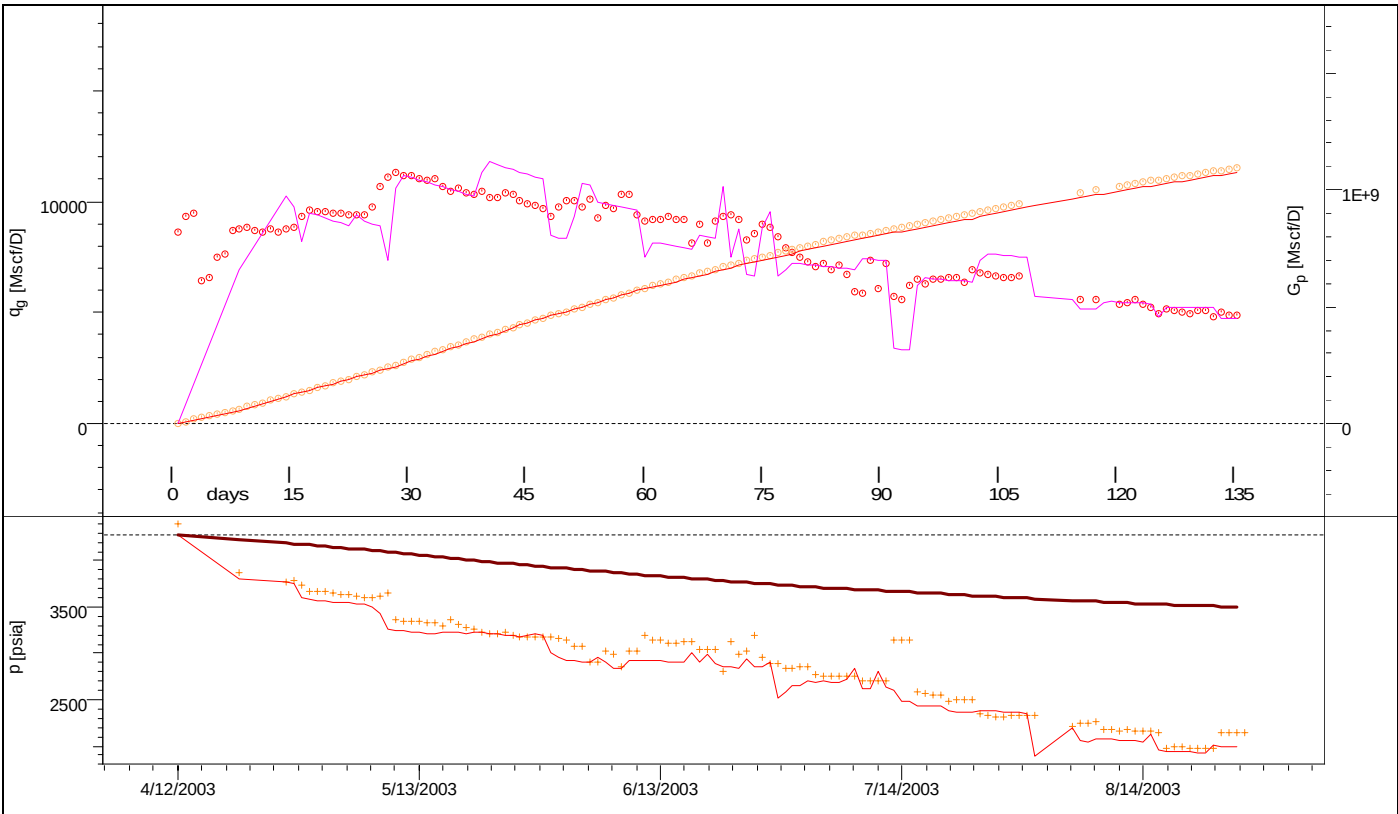


Fig 18 – 2 – Post-Acid History Match